MILITARY STANDARD

TRUCKLOADING OF HAZARDOUS MATERIALS

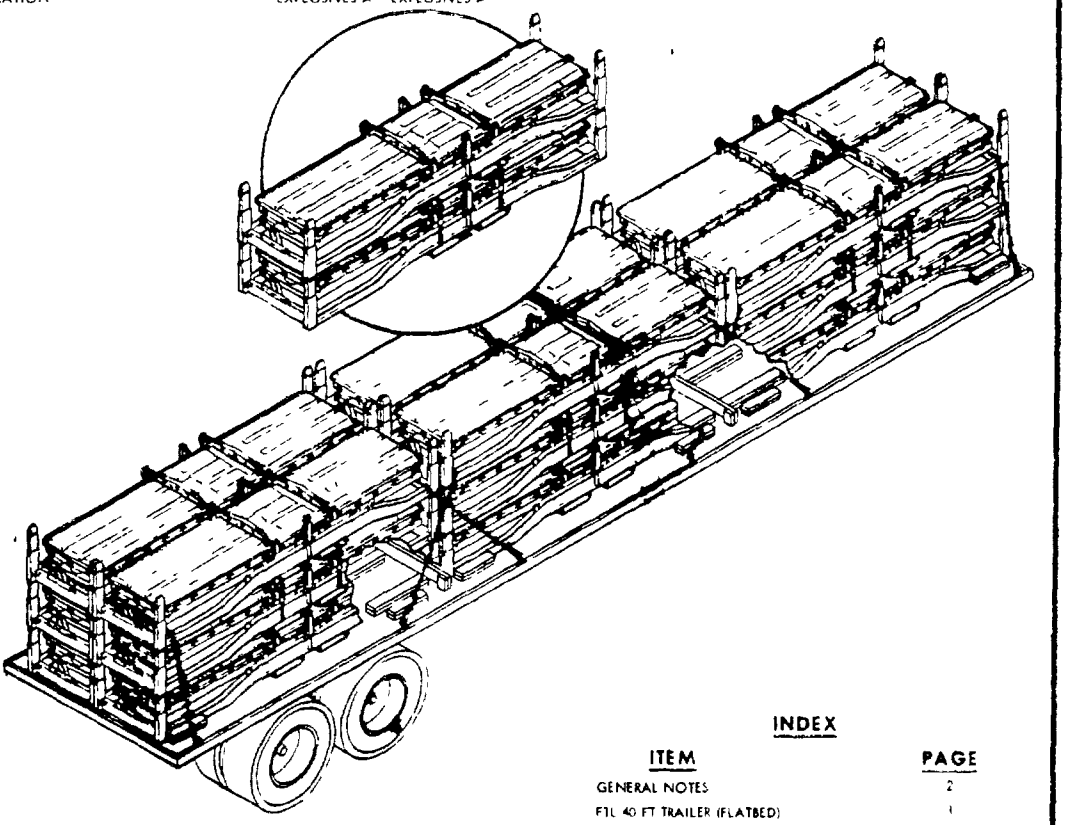
GUIDED MISSILE, AIM-7E OR 7F (SPARROW)
(LESS WINGS AND FINS)
IN CONTAINER CNU-166/E
FLEET ISSUE UNIT LOAD

MIL-STD-1320-155
(NAVY)

17 SEPTEMBER 1974

UNIT LOAD DATA

	AIM-7E	AIM-7F
UNIT LOAD DRAWING	WR-54 256	WR-54/25C
NO OF MISSILES PER CONTAINER	8	8
NO OF CONTAINERS PER UNIT LOAD	2	"
GROSS WEIGHT OF ONE CONTAINER (APPROX)	1833 LB	1972 LBS
GROSS WEIGHT OF UNIT LOAD (APPROX)	3671 LBS	3949 LBS
CUBE 162 3 CU FT		
DIMENSIONS L 159 x W 36 x H (OVERALL) 26 INCHES		
(STACKING 23 INCHES		
DOT HAZARD CLASSIFICATION	EXPLOSIVES A	EXPLOSIVES A



NOTES

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES
4. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING CONTAINERLOADING AND CARLOADING MILITARY STANDARDS REFER TO INDEX TO STANDARDS, MIL-HDBK-236 (NAVY)

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LTL 40 FT TRAILER (FLATBED)	4, 5 6
STACKING AND STRAPPING	7

FSC 8140

AUTHORIZED AND RELEASED FOR HIGHWAY SHIPMENTS ONLY

M S Gray NWHL 9/13/74
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J E Kelly 9/13/74
SIGNATURE AIRSYSCOM BY DIRECTION DATE

ORIGINATOR Charles Van Buskirk 9/13 74
SIGNATURE

NAVAL WEAPONS HANDLING LABORATORY
NAD EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR TRUCKLOADING THE GUIDED MISSILES AIM-7E OR AIM-7F (SPARROW) IN THE CNU-166/E CONTAINER
2. PROCEDURES AND PRACTICES DESCRIBED HEREIN ARE INTENDED FOR 40 FT FLATBED TRAILERS WITH THE TRAILERS TANDEM AXLES LOCATED IN THE "WESTERN POSITION" (AT THE EXTREME REAR OF THE TRAILER). DO NOT USE TRAILERS WITH METAL FLOORS.
3. THE CONTAINERS ARE UNITIZED FORMING THE FLEET ISSUE UNIT LOAD WR-54/256.
4. TO INCREASE THE TOTAL NUMBER OF CONTAINERS IN A TRUCKLOAD ONE ADDITIONAL CONTAINER IS STRAPPED TO THE FLEET ISSUE UNIT LOAD AS DESCRIBED ON PAGE 7.

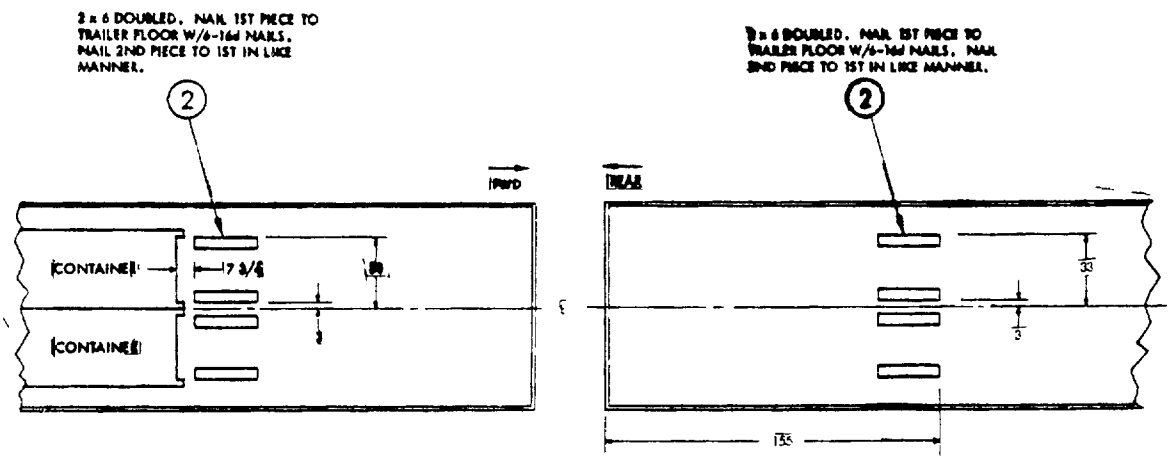
NOTE

WHEN THE TRUCK TRANSPORTATION CYCLE IS COMPLETED, THE TOP CONTAINER IS REMOVED BY CUTTING THE TOP STRAPPING. THE LOOSE CONTAINERS SHALL BE UNITIZED PER WR-54/256 FORMING A FLEET ISSUE UNIT LOAD.

5. BLOCKING (ITEM 2) MUST BE PREPOSITIONED ON THE TRAILER AS SHOWN IN DETAILS A AND B.
6. ONLY CHAINS AND LOAD BINDERS SHALL BE USED TO SECURE THE CONTAINERS TO THE TRAILER AND THEY SHALL MEET THE FOLLOWING CONDITIONS:
   (a) CHAIN AND FITTINGS SHALL BE 3/8 INCH AND LOAD BINDERS 3/8 INCH CAPACITY.
   (b) CHAINS, FITTINGS AND LOAD BINDERS SHALL HAVE A MINIMUM SAFE WORKING LOAD OF 5000 LBS.
   (c) LOAD BINDERS SHALL BE SAFETY WIRED WITH 16 GAUGE SOFT ANNEALED IRON WIRE OR SECURED USING THE END PIECE OF THE 3/8 CHAIN.
7. PRIOR TO LOADING THE TRAILER AND DURING THE PRELOADING INSPECTION REQUIRED BY NAVORD OP 2165 AND REPORTED ON DD FORM 626, THE CHAINS, FITTINGS AND LOAD BINDERS SHALL BE INSPECTED FOR STRETCH, GOUGING, BENT LINKS, WEAR AND ANY OTHER NOTICEABLE DEFECTS. THE INSPECTOR SHALL CONFIRM THAT THE CHAINS AND LOAD BINDERS HAVE BEEN INSPECTED AND SHALL SO NOTE IN ITEM NO. 27 OF DD FORM 626. ANY DEFICIENCY SHALL BE CAUSE FOR REJECTION OF A CHAIN OR LOAD BINDER.
8. THE MAXIMUM GROSS WEIGHT OF THE TRACTOR-TRAILER AND THE ALLOWABLE AXLE WEIGHTS ARE THE RESPONSIBILITY OF THE CARRIER. THE CARRIER WILL ADVISE THE SHIPPER OF THESE LIMITATIONS AND THE SHIPPER SHALL LOAD THE TRAILER IN SUCH A MANNER THAT THE TRACTOR-TRAILER WILL NOT EXCEED THESE LIMITATIONS.
9. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS TO AN ACCESSIBLE AREA AND ATTACH "EXPLOSIVES A" PLACARD TO BOTH SIDES, FRONT AND BACK OF TRAILER.
10. APPLICABLE MATERIAL SPECIFICATIONS: DUNNAGE LUMBER, MM L-751; NAILS, FF-N-105, TYPE II, STYLE 10, COMMON BRIGH
11. FOR GENERAL TRUCKLOADING PROCEDURES REFER TO THE GENERAL TRUCKLOADING DOCUMENT, MIL-STD-1320

FTL 40 FT TRAILER (FLATBED)

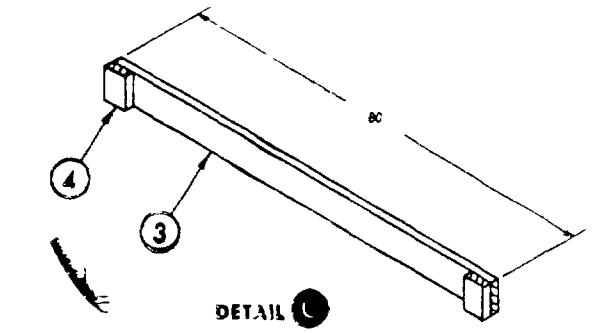
1. A FULL TRAILER LOAD CONSISTS OF 18 CNU-166/E CONTAINERS OF AIM-7E OR AIM-7F GUIDED MISSILES.



DETAIL A
LOCATION OF FORWARD PREPOSITIONED BLOCKING

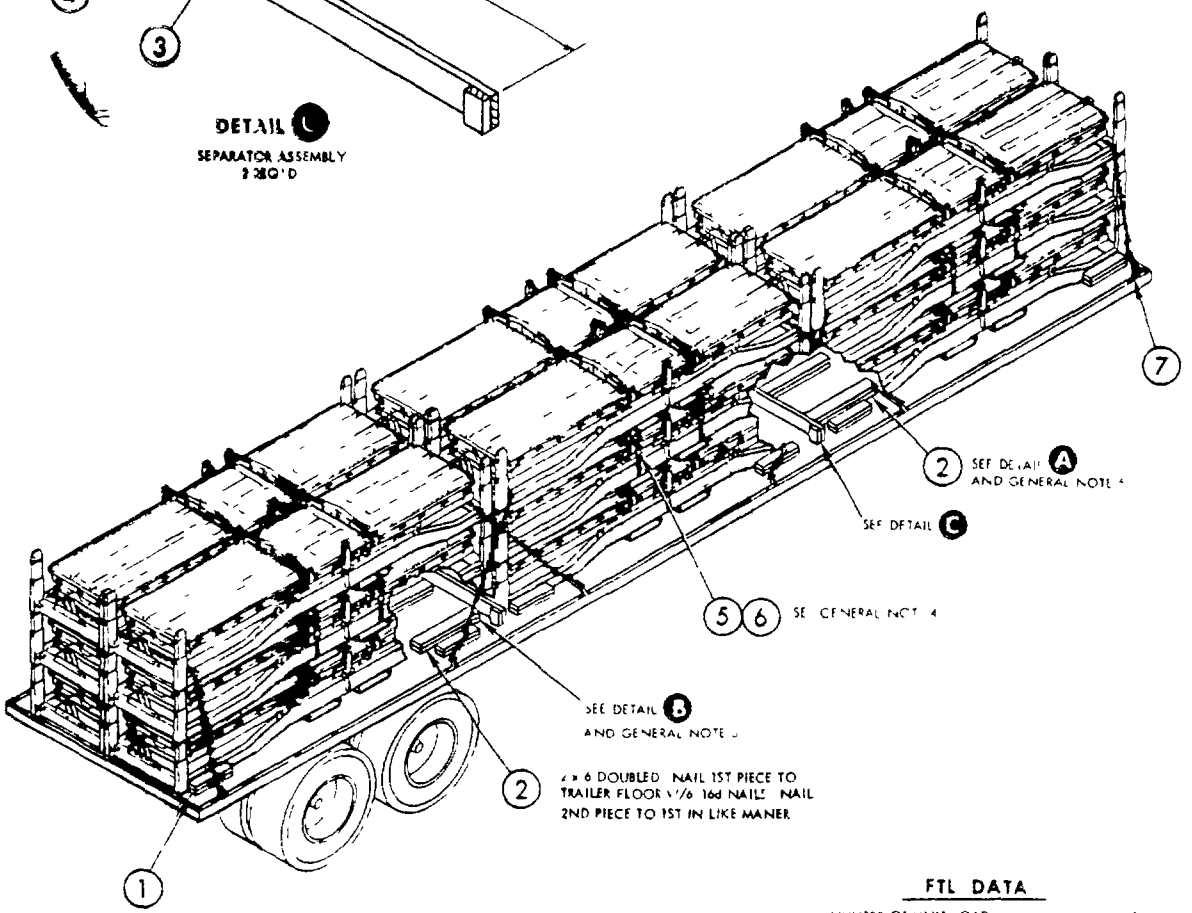
DETAIL B
LOCATION OF AFT PREPOSITIONED BLOCKING

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DETAIL C
SEPARATOR ASSEMBLY
2 REQ'D

NOTE
TRAILER SHALL BE LOADED STARTING AT THE REAR



FTL DATA

NUMBER OF UNIT LOAD	4
NUMBER OF CONTAINER	16
WEIGHT (MAX)	30,000 LBS
WEIGHT (MIN)	00,00 LB

NOTE
UNIT LOAD STRAP OMITTED FOR CLARITY

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
	LIS OF MATERIALS AND NAILING DATA					
7	CHAIR & LOAD BINDER	10 INCH	4			
6	UNITIZING SEAL	1 4	12			
5	UNITIZING STRAPPING	1 4 035 12 FT	12		-	
4	CLEAT	4 x	4			
3	CROSSMEMBER	x 6	4			10d
2	BLOCKING PREPOSITIONED	2 6 x 8	6	SEE DETAIL A & B		
1	SLEEPER	2 x 6	24	SEE FIELD NOTE		

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LTL 40 FT TRAILER (FLATBED)

WHEN REQUIRED TO SHIP LESS-THAN-TRUCKLOAD SELECT THE CORRECT LOAD PATTERN FOR THE NUMBER OF UNIT LOADS TO BE SHIPPED FROM TABLE I AND POSITION THE UNIT LOADS ON THE TRAILER ACCORDINGLY

2 WHEN BLOCKING THE FRONT AND REAR OF 5 UNIT LOADS OR LESS, USE 4 x 6 x 30 INCH BACK UP CLEATS

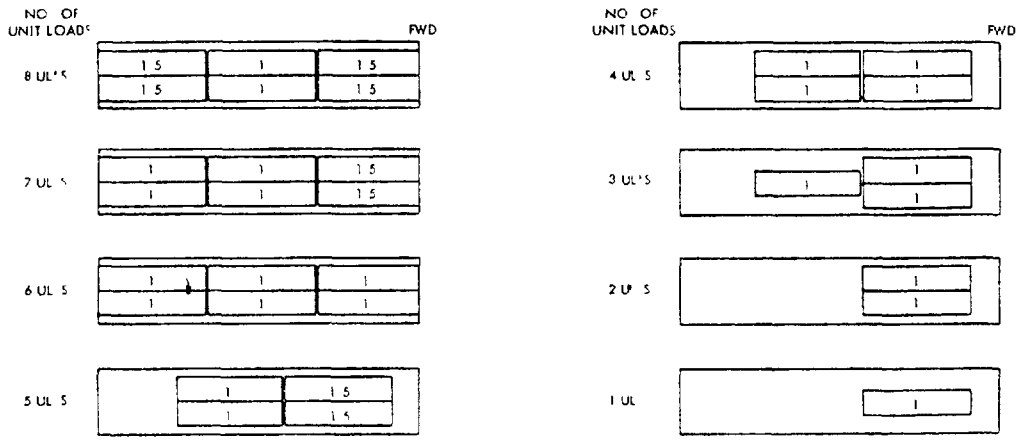
3 WHEN THE LOAD PLAN REQUIRES THE STACK TO BE 1 5 UNIT LOADS HIGH (THREE CONTAINERS HIGH) THEY SHALL BE STRAPPED TOGETHER AS SHOWN ON PAGE 7

4 THE LTL SHOWN ON PAGE 5 IS FOR 5 UNIT LOADS AND THE LTL LOAD SHOWN ON PAGE 6 IS FOR 3 UNIT LOADS THIS DOES NOT LIMIT LTL SHIPMENTS TO THIS QUANTITY SHIPMENTS MAY CONSIST OF A GREATER OR LESSER NUMBER OF UNIT LOADS

5 ACTUAL SHIPPING REQUIREMENTS WILL DETERMINE THE NUMBER OF UNIT LOADS TO BE SHIPPED THE LOAD PATTERN, BLOCKING, BRACING AND TIE DOWNS SHALL BE IN ACCORDANCE WITH THE APPROPRIATE PARTS OF THIS MILITARY STANDARD AND THE GENERAL TRUCKLOADING DOCUMENT MIL-STD-1320 (NAVY)

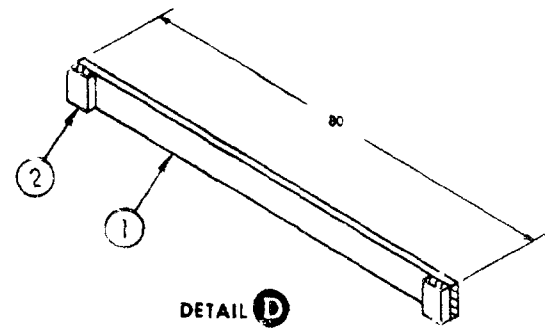
TABLE I

LOAD PATTERNS *

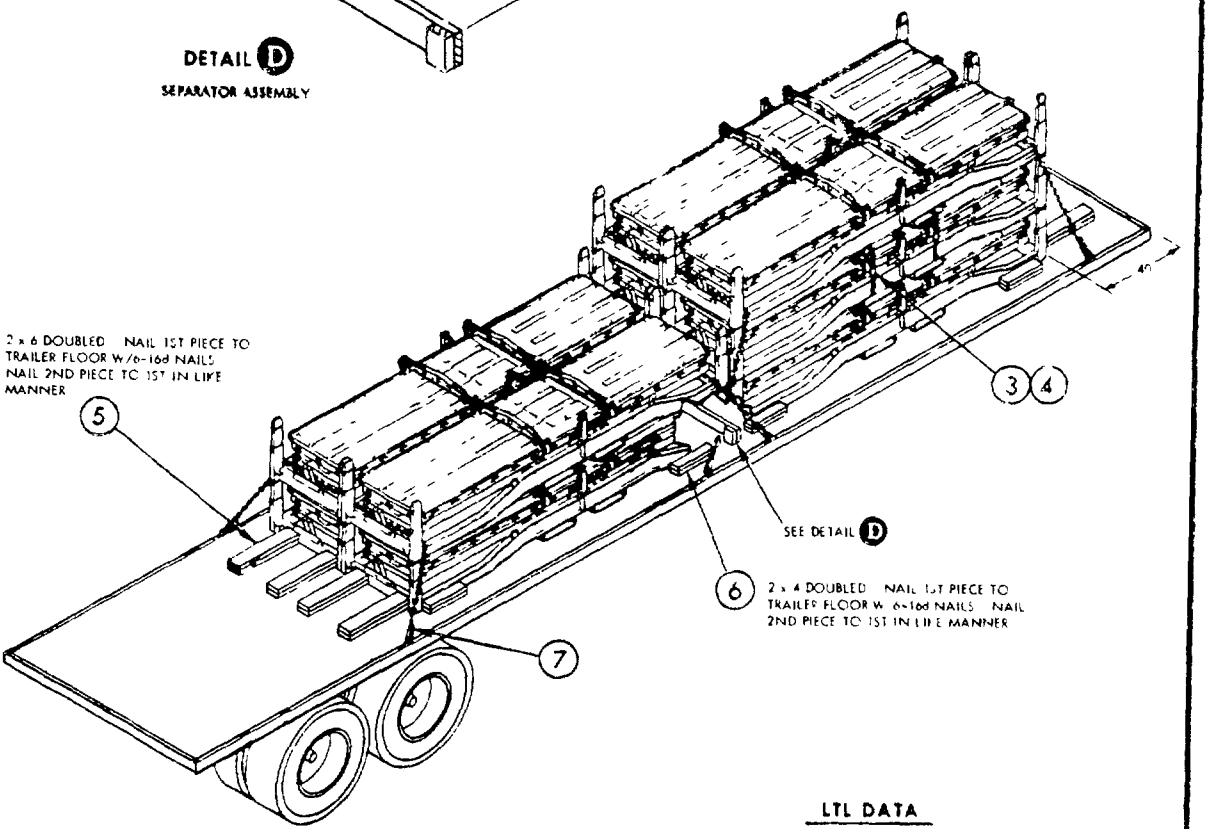


*LOAD PATTERN IS THE FLOOR PLAN FOR THE LOAD THE NUMBER IN THE BLOCKS IS THE NUMBER OF LAYERS OF UNIT LOADS IN THE STACK THE NUMBER 1 5 INDICATES THAT THE STACK IS 1 5 UNIT LOADS HIGH (OR THREE CONTAINERS HIGH SEE GENERAL NOTE 4 AND NOTE PAGE 2.

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DETAIL D
SEPARATOR ASSEMBLY



LTL DATA

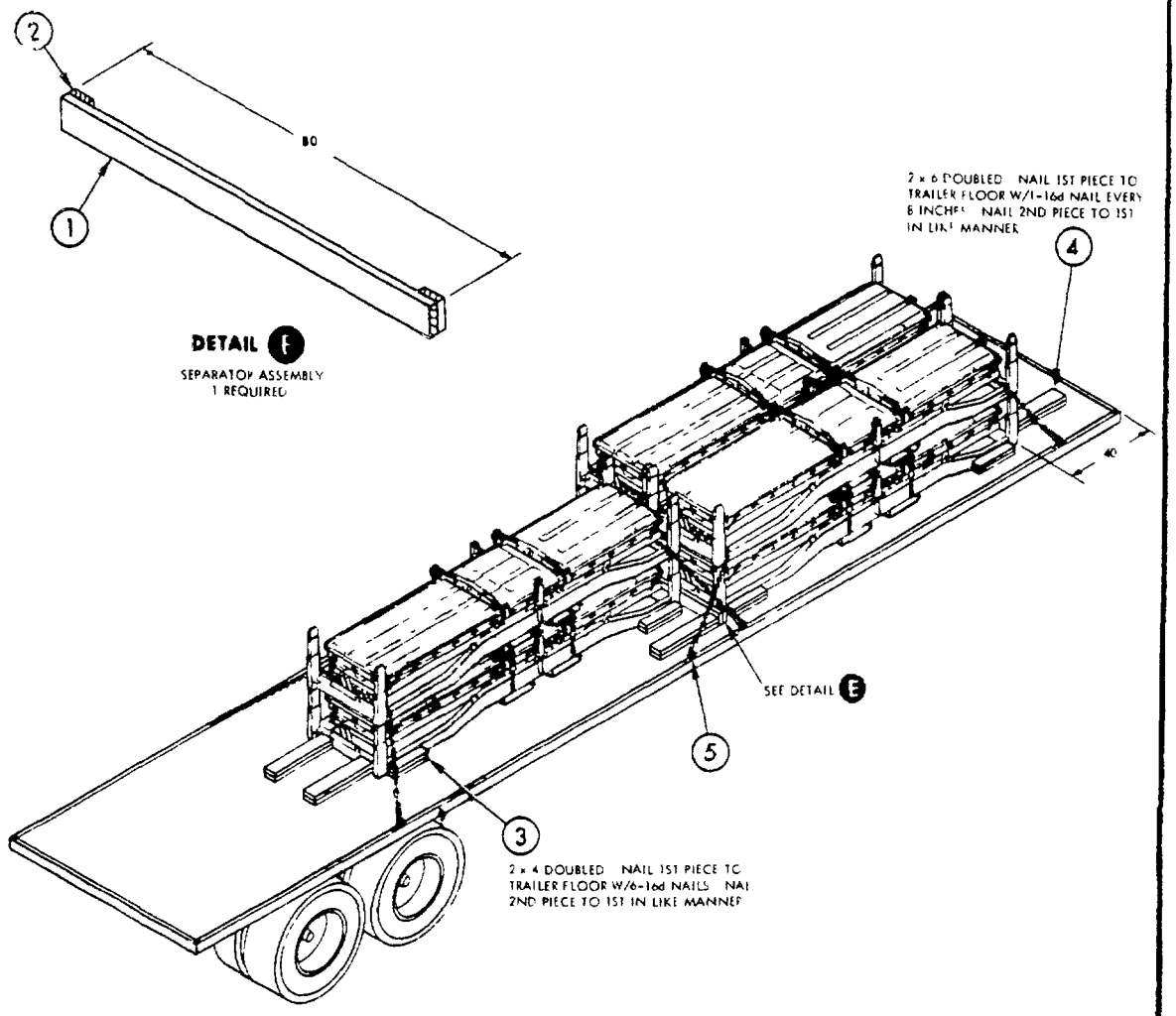
NUMBER OF UNIT LOADS	
NUMBER OF CONTAINER	1C
WEIGHT (AIM-7F	16 350 LBS
WEIGHT (AIM-7F	19 745 LBS

NOTE

UNIT LOAD STRAPS OMITTED FOR CLARITY

PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER	SIZE
	LIST OF MATERIALS AND NAILING DATA				NAILS	
7	CHAIN & LOAD BINDER	3 8 INCH	4	-		-
6	SLEEPER	4 x 4 x 18	16	SEE FIELD NOTE		
5	BACK UP CLEATS	2 x 6 x 30	16	SEE FIELD NOTE		
4	UNITIZING SEAL	1 1 4	4			-
3	UNITIZING STRAPPING	1 1 4 x .035 x 12 F	4		-	
2	CLEAT	2 x 4 x 5 1 2	4	-	-	
1	CROSSMEMPER	4 x 6 x 80	1	2	3	10d

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LTL DATA

NUMBER OF UNIT LOADS	3
NUMBER OF CONTAINERS	6
WEIGHT (AIM-7E)	11 013 LBS
WEIGHT (AIM-7F)	11 84 LBS

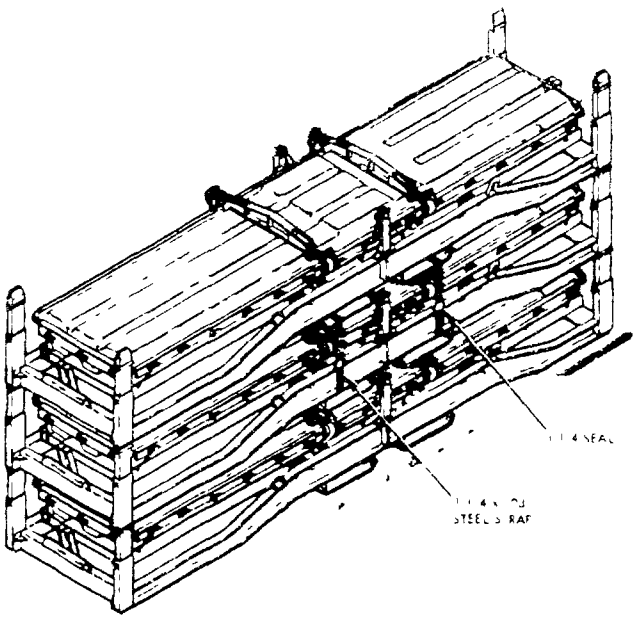
PIECE NO	DESCRIPTION	SIZE	NO PCS REQD	NAIL TO	NUMBER	SIZE
					NAILS	
5	CHAIN & LOAD BINDER	3 8 INCH	-			-
4	BACK UP CLEAT	2 x 6 x 3L	16	SEE FIELD NOTE		
3	SLEEPER	2 x 4 x 16	16	SEE FIELD NOTE		
2	CLEAT	2 x 4 x 1 2		-		
1	CROSSMEMBER	1 x 6 x 80	1	2	3	10d

LIST OF MATERIALS AND NAILING DATA

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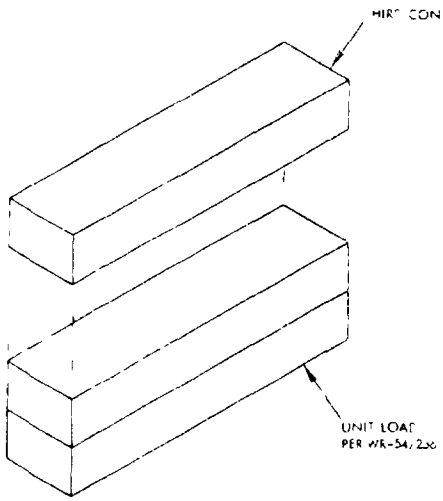
STACKING AND STRAPPING PROCEDURE

1. POSITION A THIRD CONTAINER ON TOP OF THE UNIT LOAD AS SHOWN IN DETAIL F. MAKE CERTAIN THE SUPPORT POSTS ARE PROPERLY ENGAGED.
2. POSITION STRAPPING THROUGH THE FORK POCKETS OF THE TOP CONTAINER OF THE UNIT LOAD AND THROUGH THE FORK POCKETS OF THE MIDDLE CONTAINER.
3. TENSION STRAPPING AND DOUBLE NOTCH SEALS.



NOTE

UNIT LOAD STRAPS OMITTED FOR CLARITY



DETAIL F

REVIEW ACTIVITIES
(NAVY - OS AS)

PREPARING ACTIVITY
NAVY - 0
PROJECT NO. 8140-N246

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS This sheet is to be filled out by personnel either Government or contractor involved in the use of the specification in procurement of products for ultimate use by the Department of Defense This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost Comments and the return of this form will be appreciated Fold on lines on reverse side staple in corner and send to preparing activity Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements

SPECIFICATION

MIL-STD-1320-155 (NAVY)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 68 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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POSTAGE AND FEES PAID

OFFICIAL BUSINESS

Commanding Officer
Naval Ammunition Depot Earle
Naval Weapons Handling Laboratory
Colts Neck, New Jersey 07722

Attn: 803

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